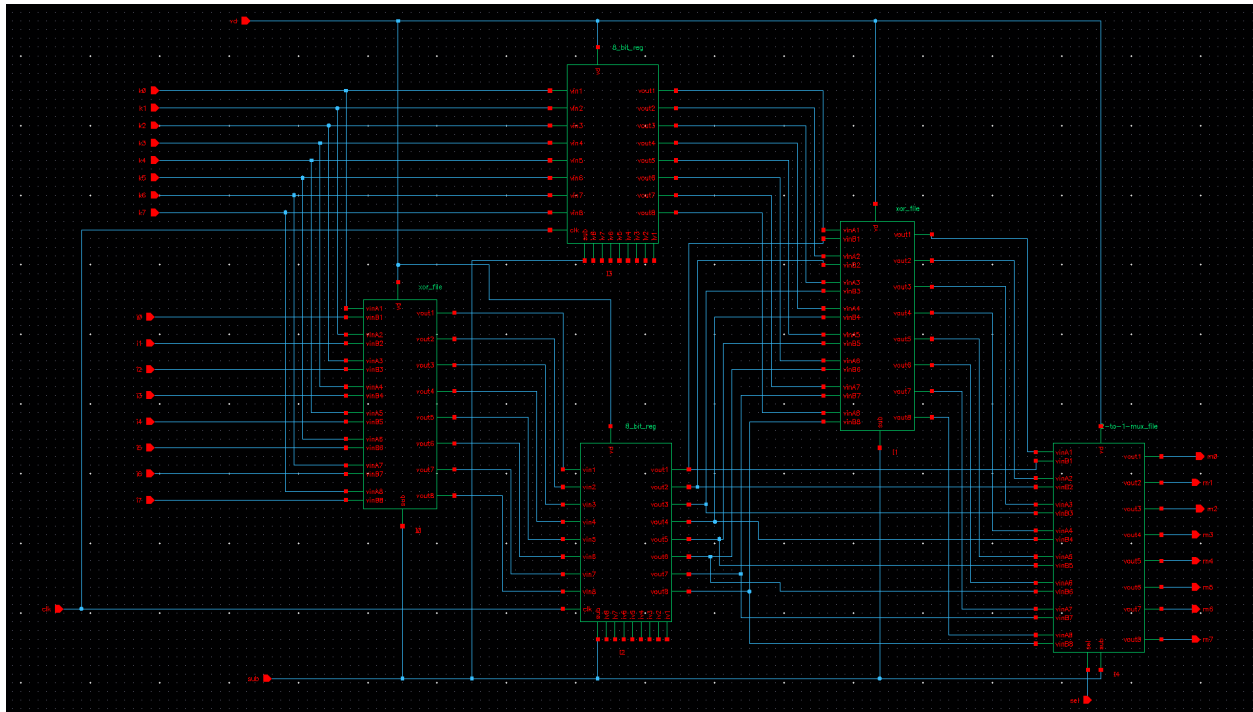


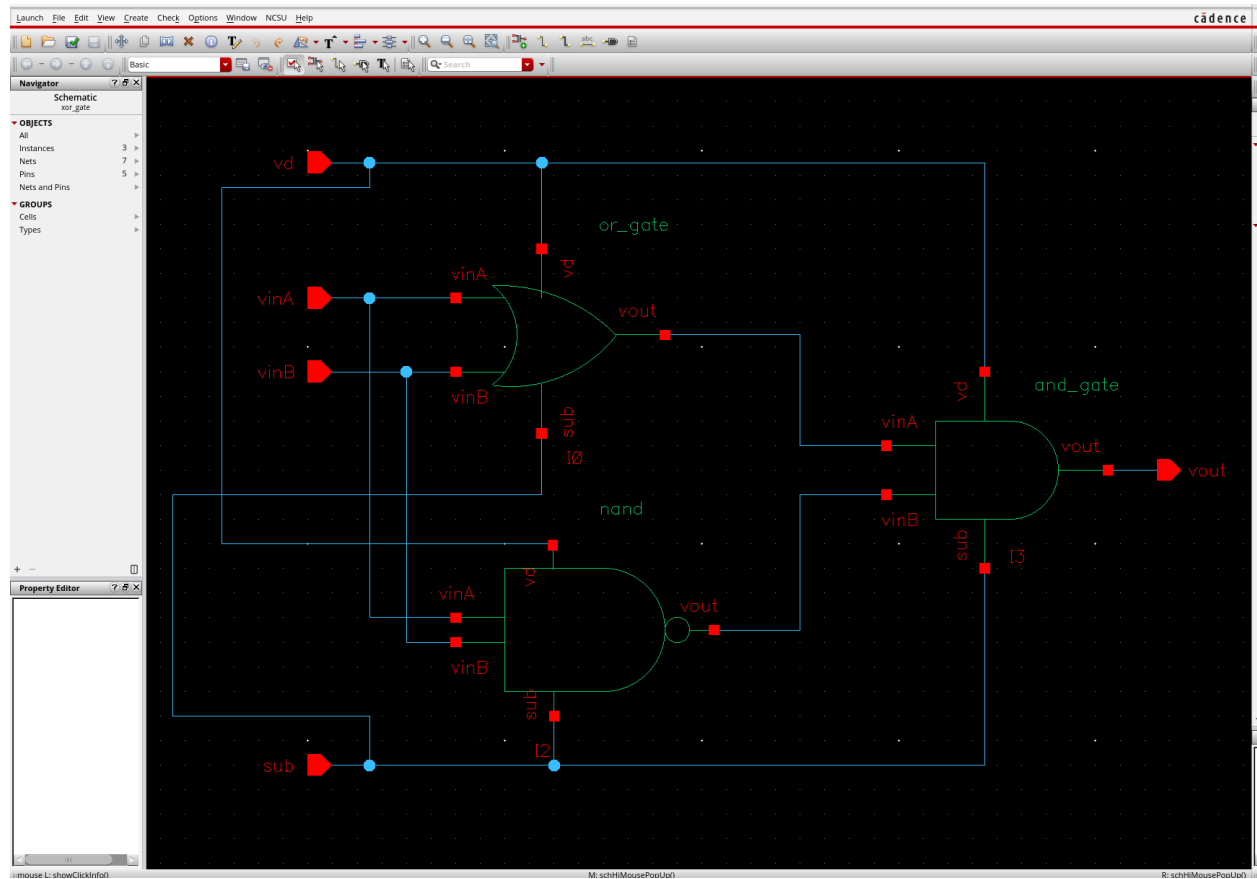
XOR Encryptor/ Decryptor
Project 2 -Schematic and Sim
Gerald Stanford



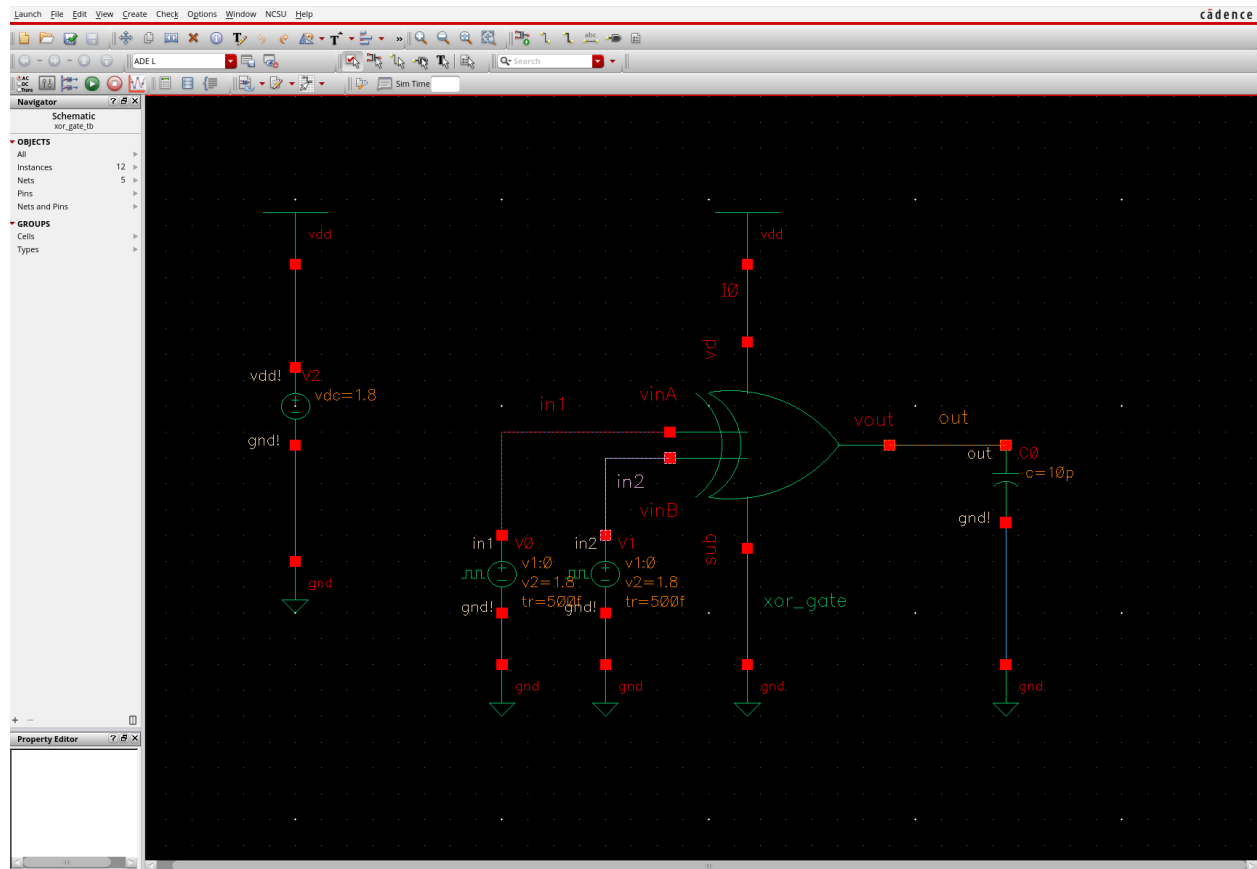
Schematic - 3 sections

- Encryption
- Storage
- Decryption

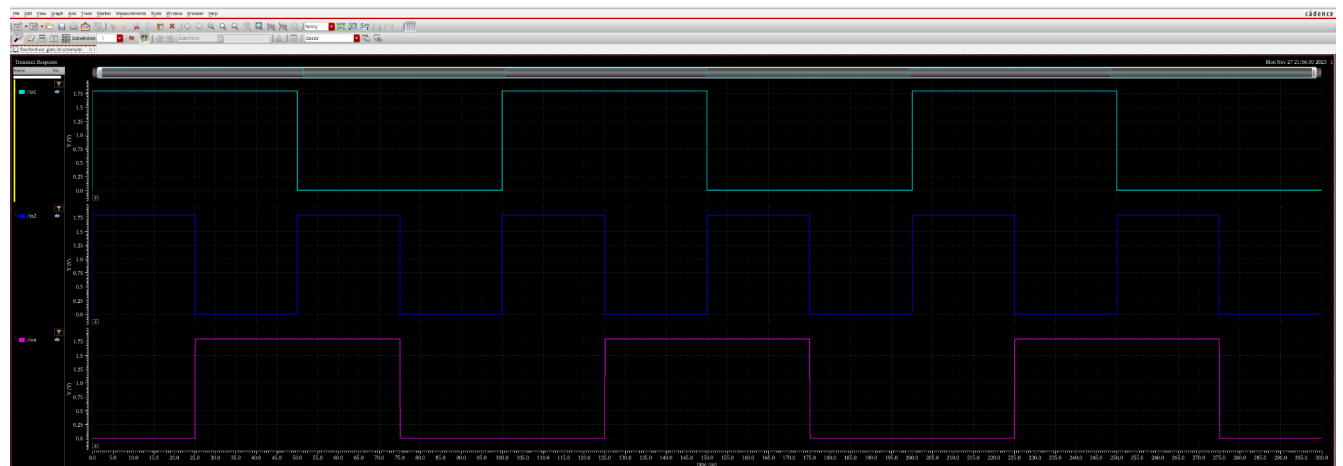
Encryption:



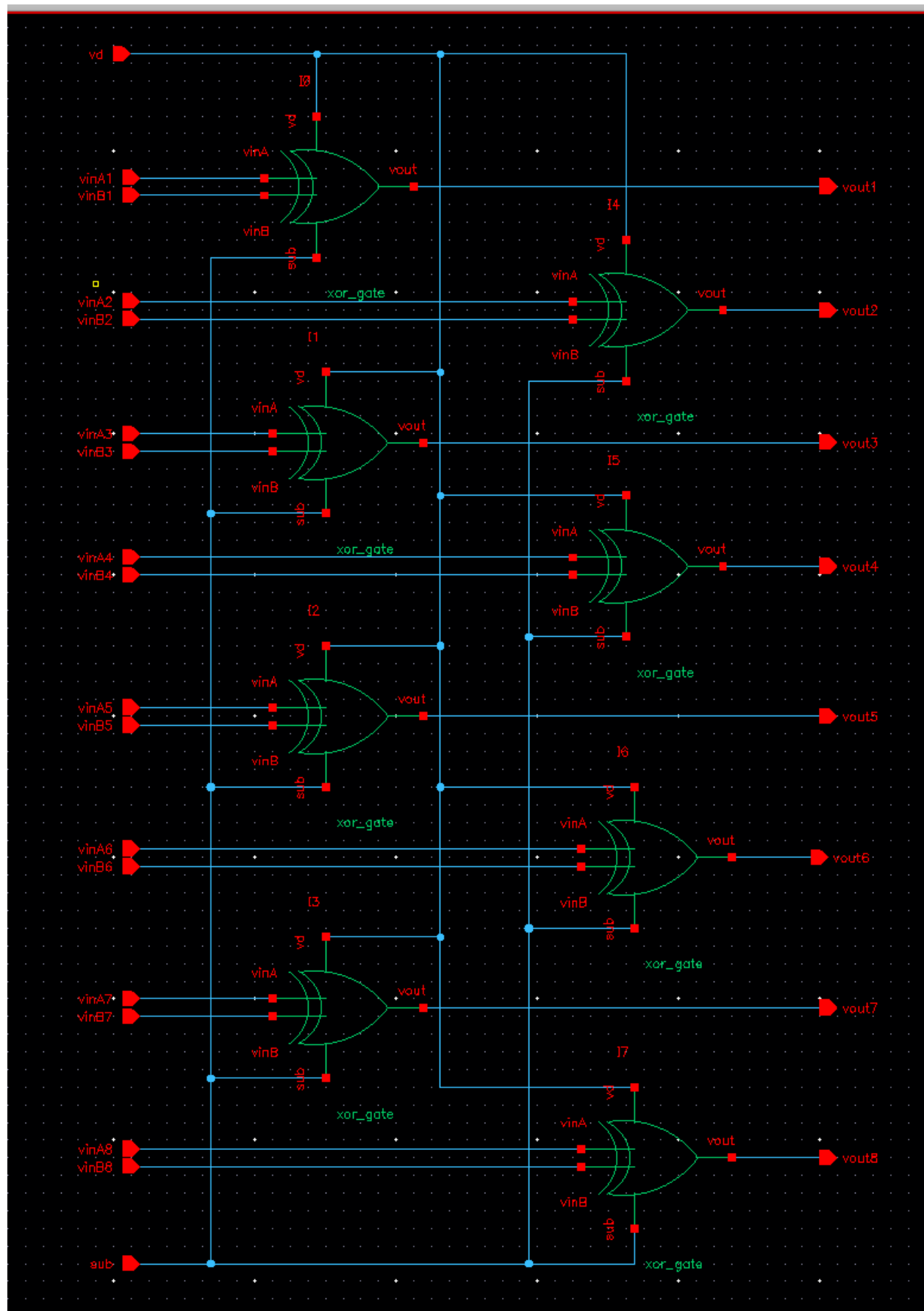
Xor - gate

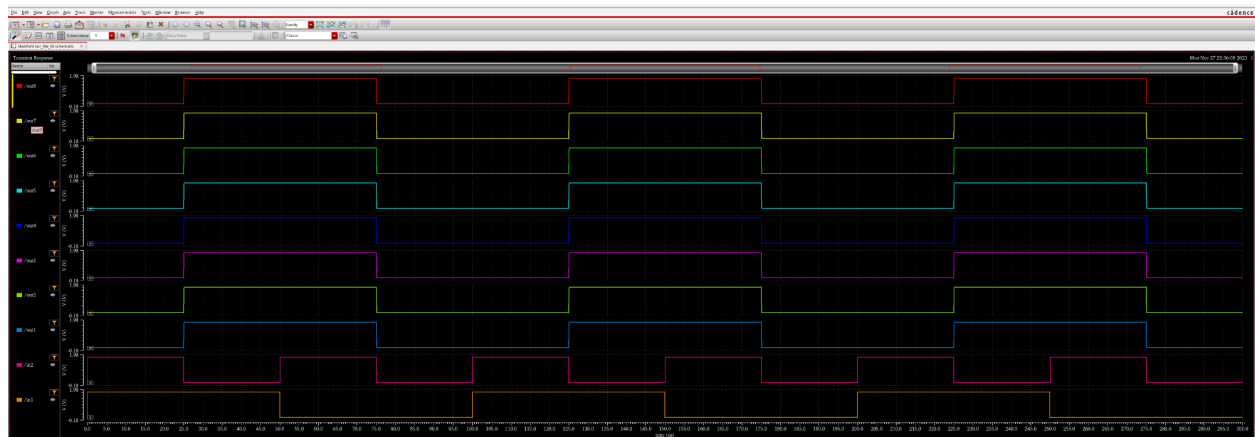
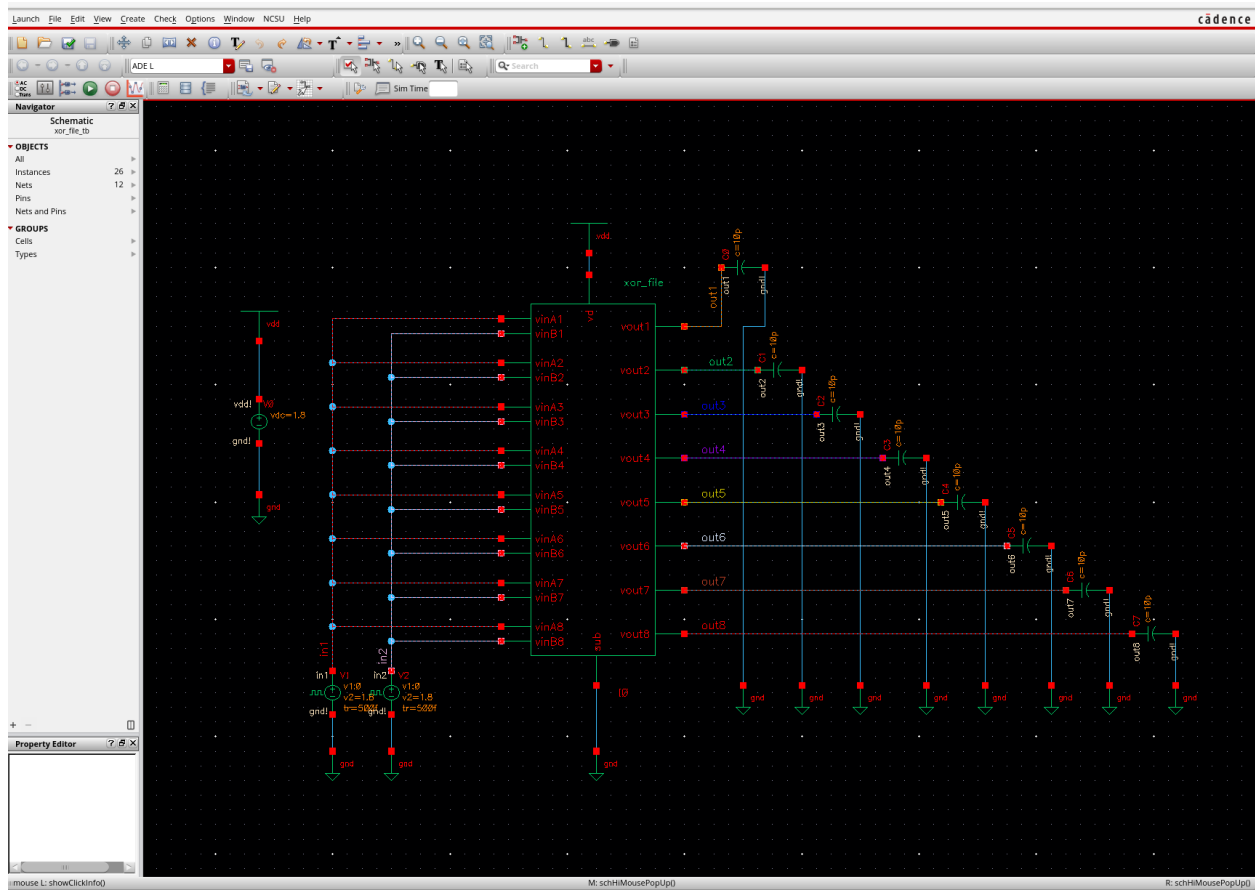


Xor - Gate Test Bench



Xor gate - Sim
Shows the 2 input Xor Gate truth table

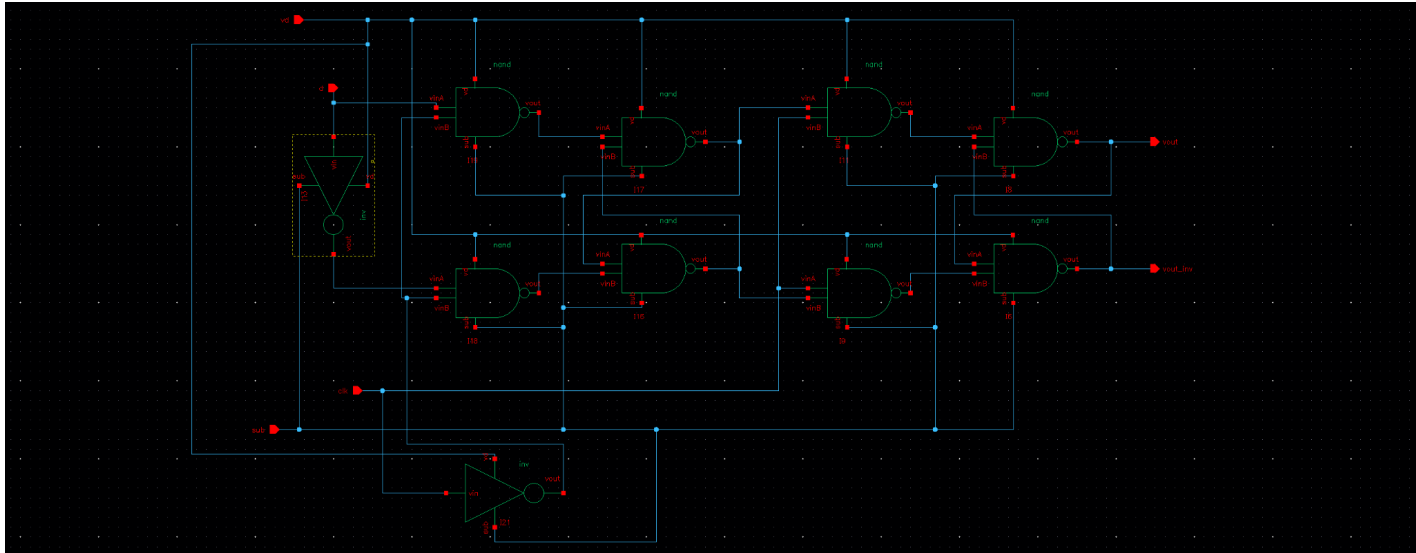




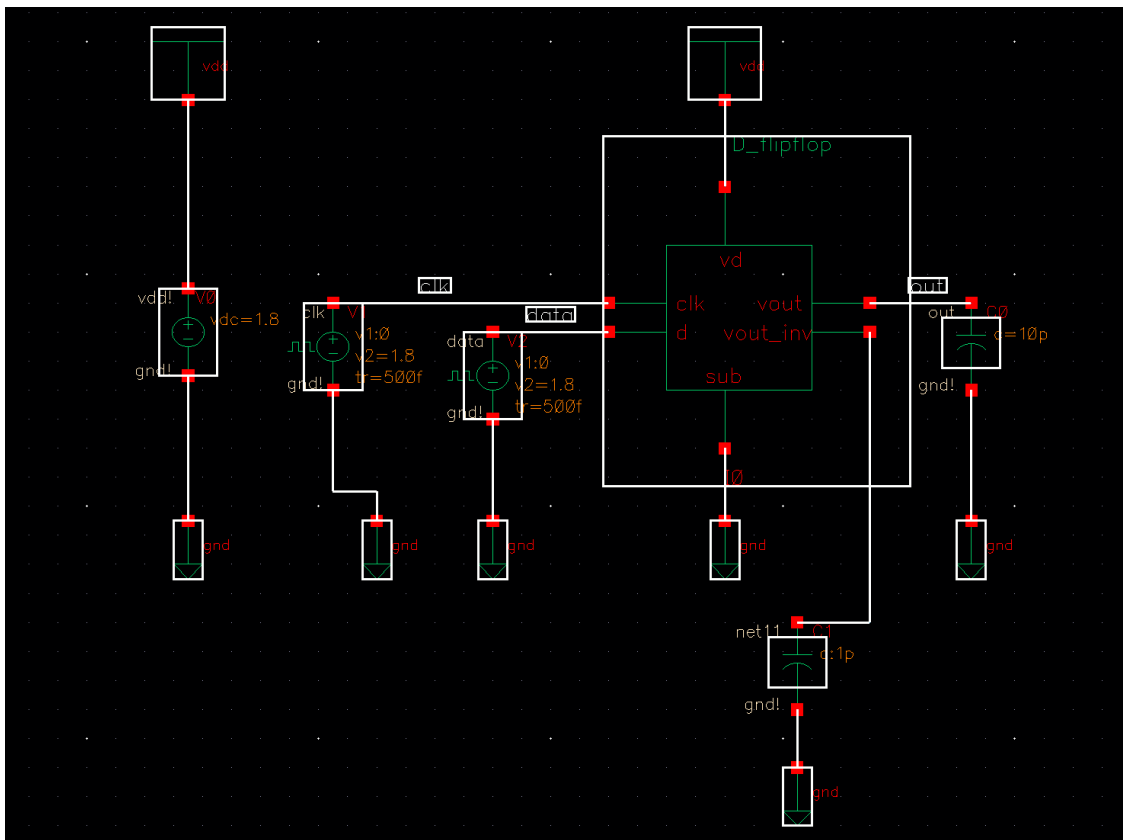
Xor File - Schematic/ TestBench/ Sim

The Xor file is just 8 Xor gates in parallel. The Sim is identical to the xor gate sim

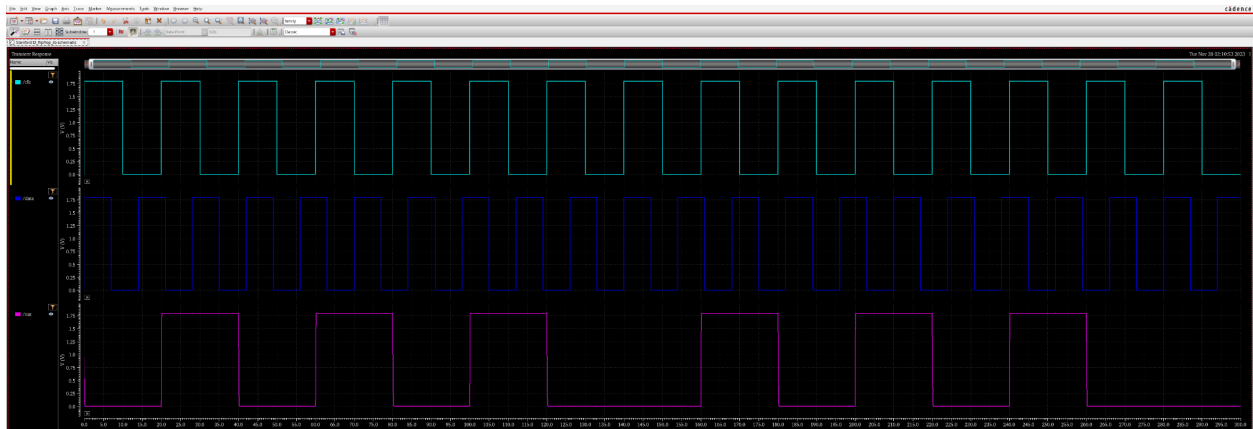
Storage:



D- Flip Flop schematic



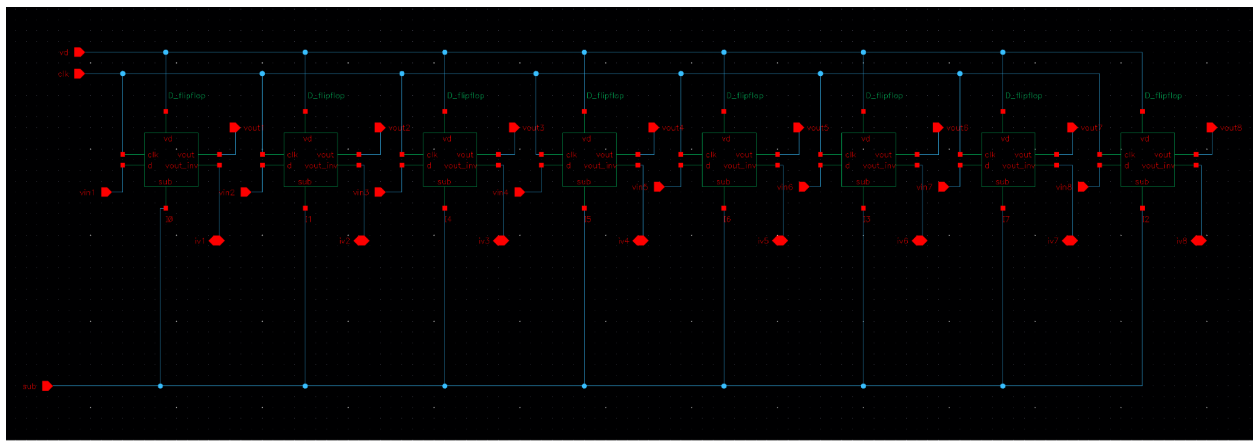
D- Flip Flop Testbench schematic

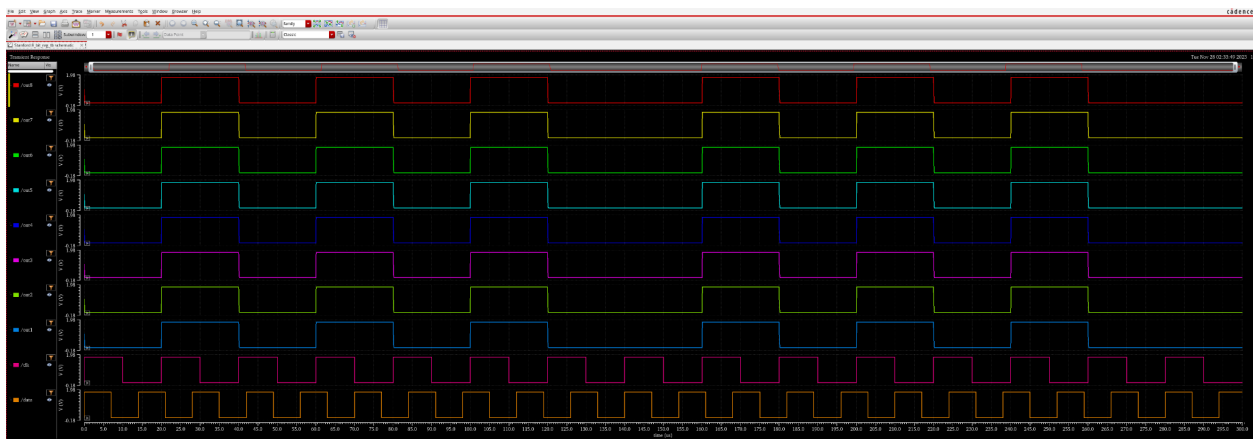
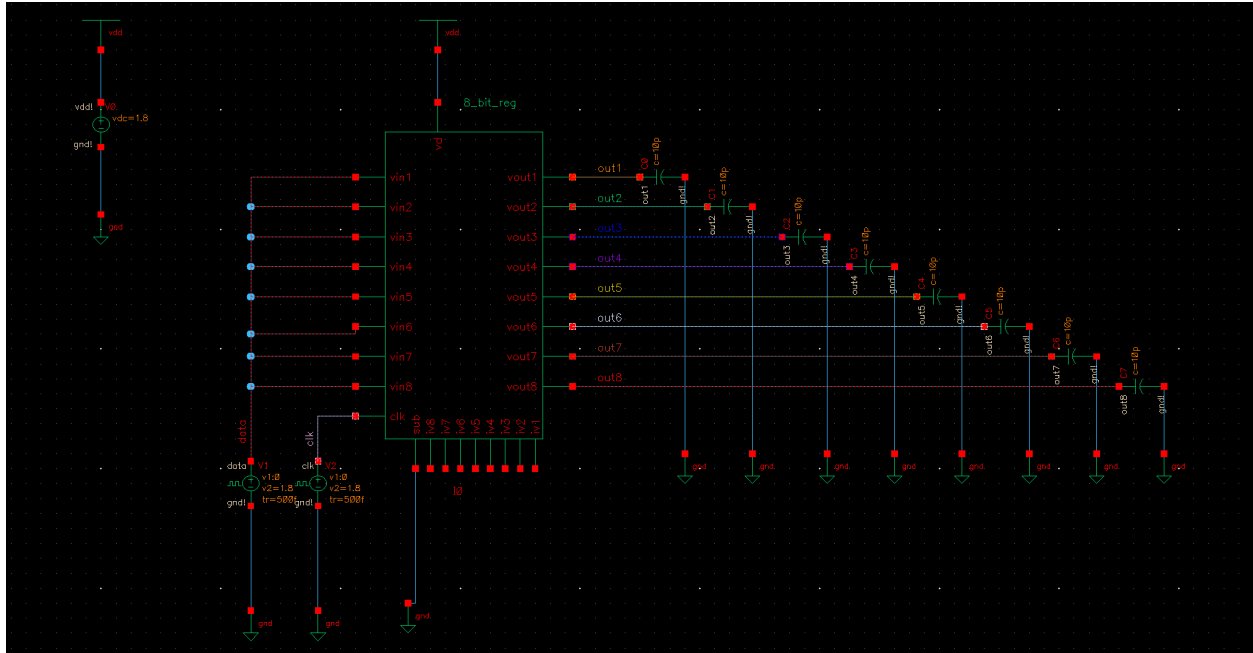


D- Flip Flop Testbench sim

Since the Flip Flop is edge triggered, the data output at the edge is what is output.

Sections where the Data and Clk rise at the same time output 0



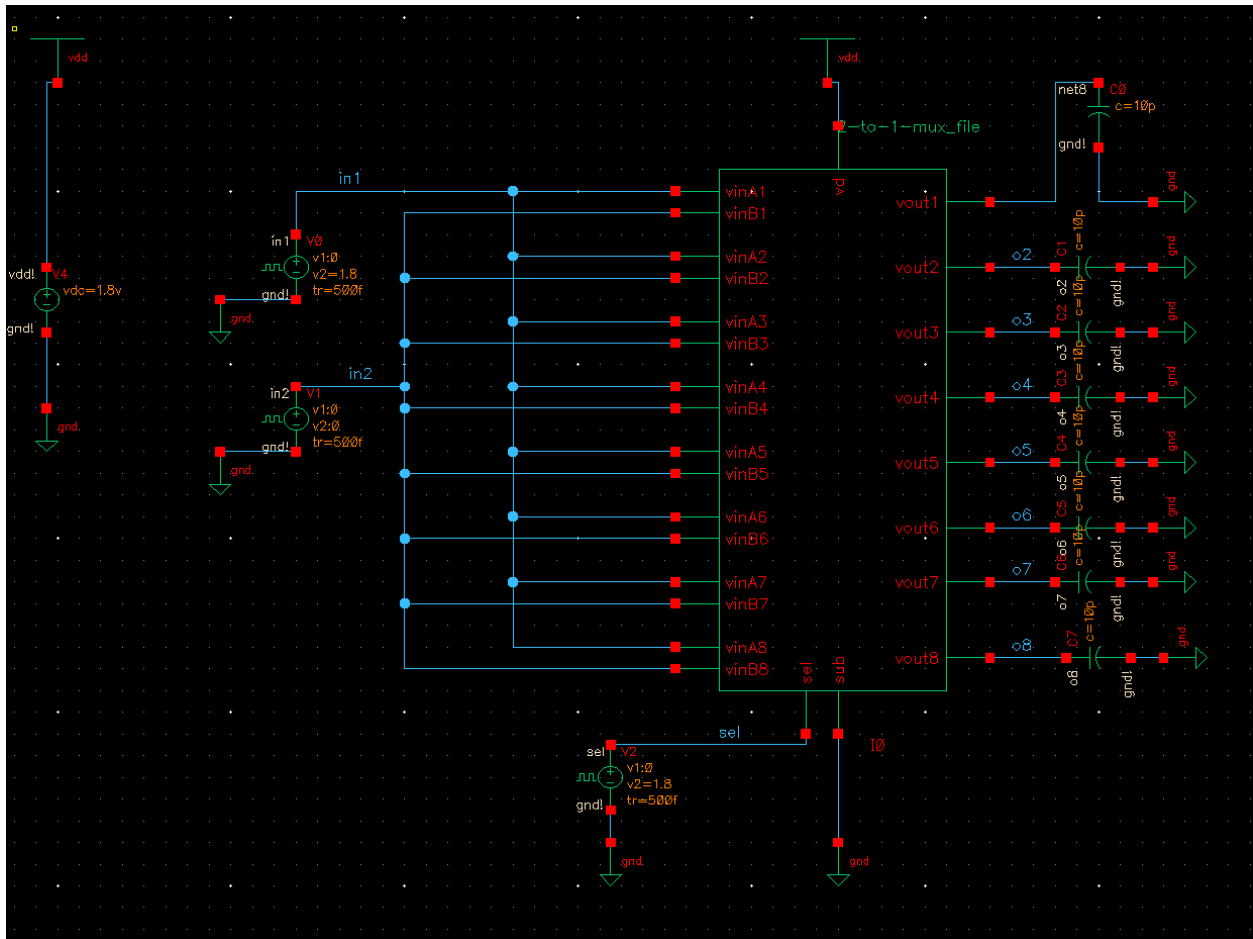
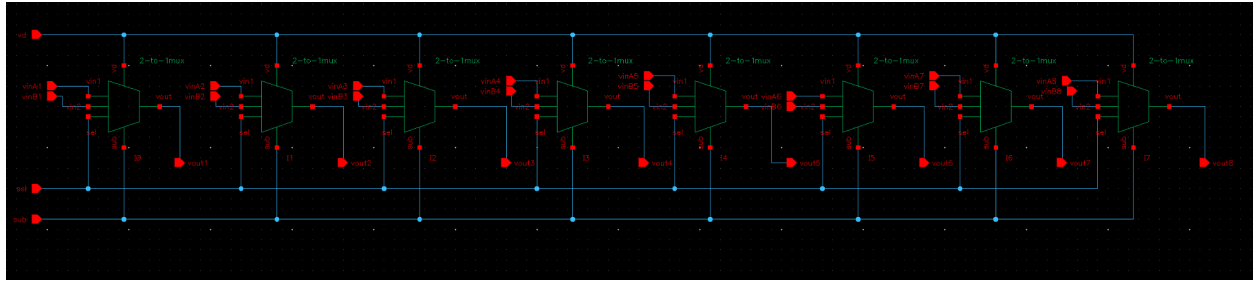


8 bit register - Schematic / TestBench / Sim

Like the Xor file, the 8 bit register is just 8 parallel flip flops, so the sim is identical to the single version.

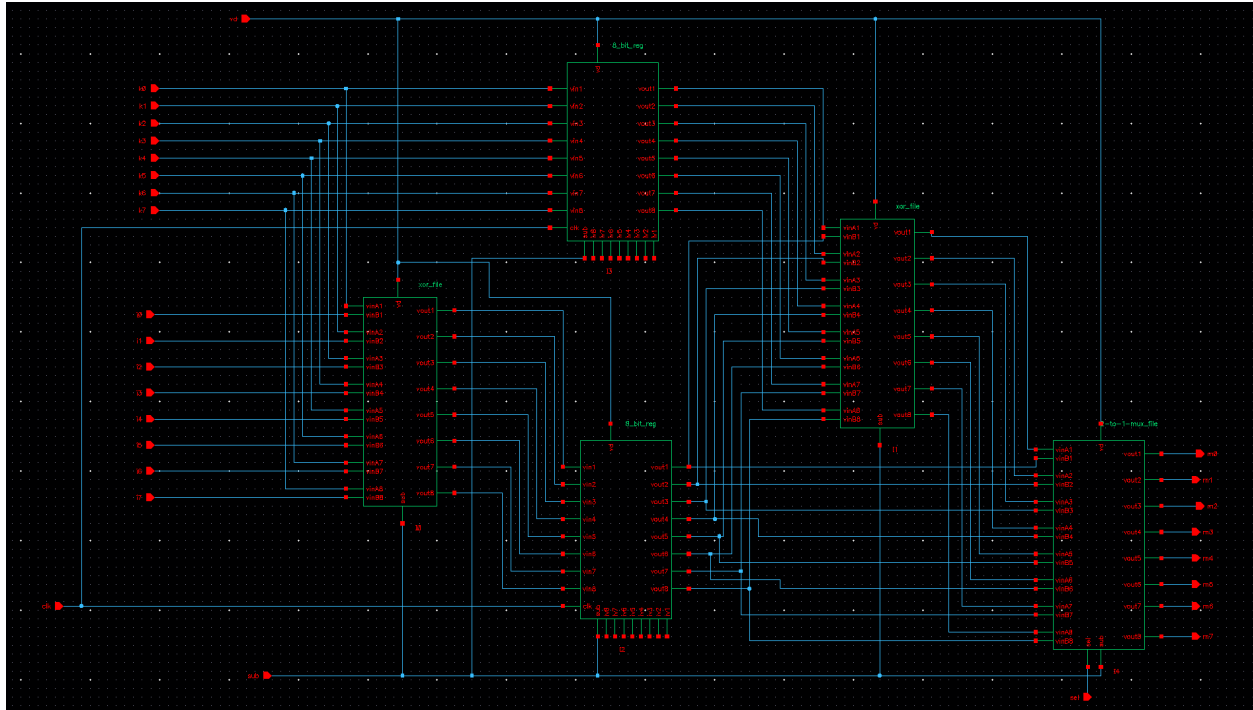
Decryption:

Decryption uses another xor file and a 2-to-1 mux file for optional decryption.



Mux-file - Schematic and Testbench Schematic.

Full Encryptor:

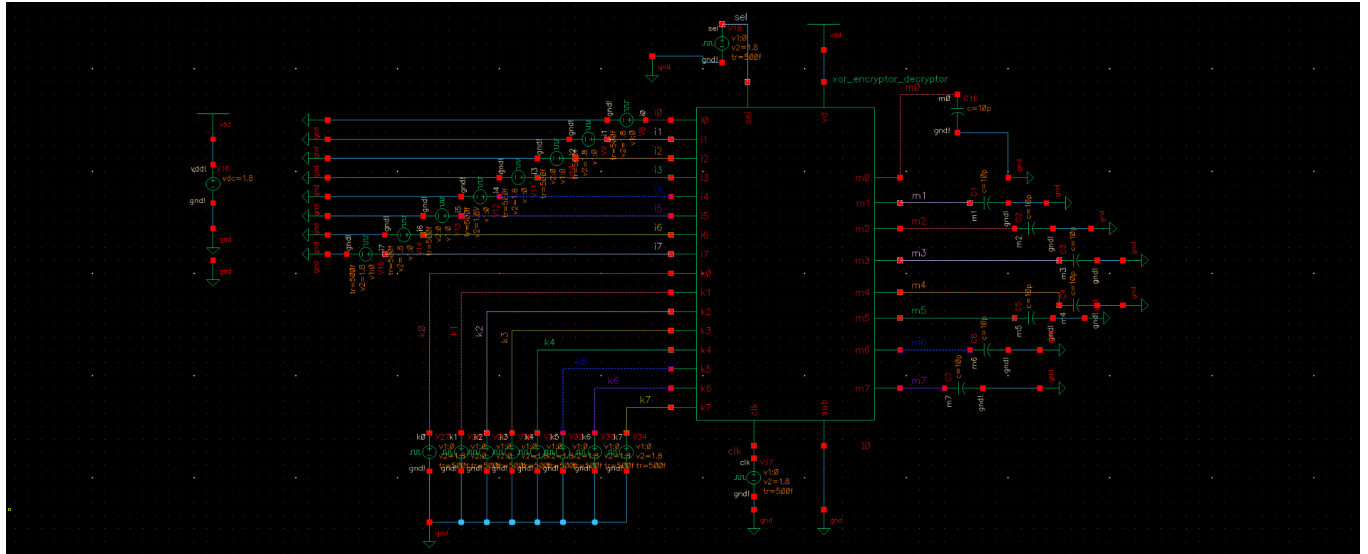


The Key inputs are at the top and are sent to the first xor file for encryption and the top most 8 bit register for storage.

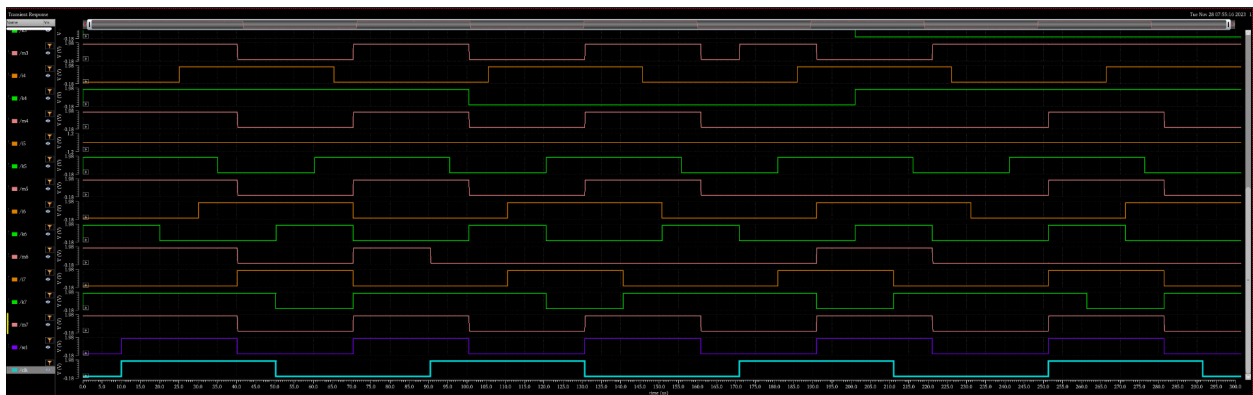
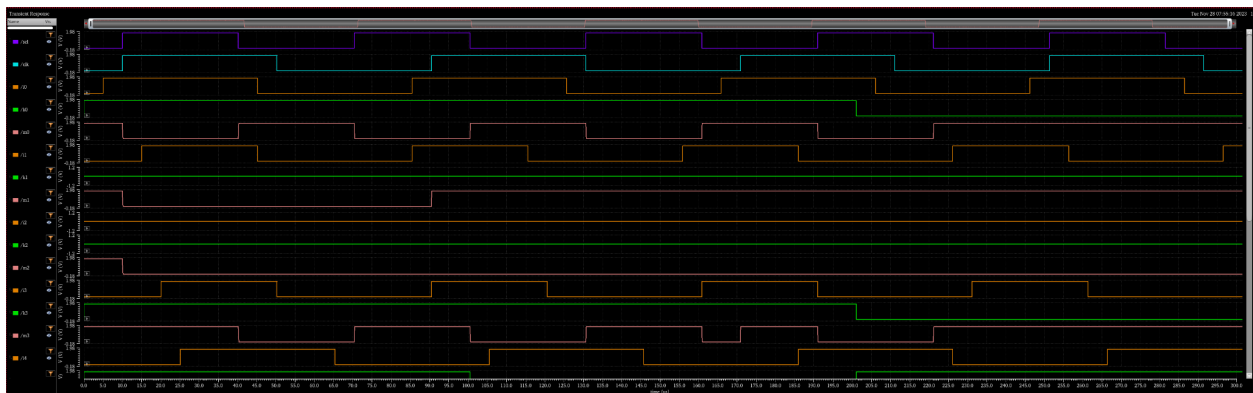
The data inputs are lower and head into the xor file.

The encrypted data and key are stored in their own registers.

For decryption, the encrypted data and the same key are passed to the second xor file. Using the mux file, either the encrypted data or original version can be output.



Encryptor Testbench



Encryptor - Sim results